Breast Cancer Biopsy Analysis | Statistical Modeling & Data Visualization

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Dataset

- We chose the biopsy dataset because it has the most amount of data to work with.
- Contains multiple numerical values for biopsy samples (benign or malignant).
    - Gives us a defined outcome to source hypothesis from.
- Multiple different features and variables to compare to each other.
- Allows us to use statistical methods to see what variables are strongly correlated to benign or malignant.
- Comparison of different biopsy characteristics
    - Ex. clump thickness or cell shape compare and what is the outcome on benign or malignant

Variables

V6 - Numerical, discrete, ordinal

V7 - Numerical, discrete, ordinal

V8 - Numerical, discrete, ordinal

V9 - Numerical, discrete, ordinal

Class - Categorical, discrete, nominal

Variable Descriptive Statistics

Variable 1 (Clump Thickness):

- Mean: 4.418
- Median: 4
- Mode: 1
- Variance: 7.928
- Standard deviation: 2.816
- Skewed distribution? - Yes (positive)

Variable 2 (Uniformity of Cell Size):

- Mean: 3.134
- Median: 1
- Mode: 1
- Variance: 9.311
- Standard deviation: 3.051
- Skewed distribution? - Yes (positive)

Variable 3 (Uniformity of Cell Shape):

- Mean: 3.207
- Median: 1
- Mode: 1
- Variance: 8.832
- Standard deviation: 2.972
- Skewed distribution? - Yes (positive)

Variable Descriptive Statistics Cont.

Variable 4 (Marginal Adhesion):

- Mean: 2.807
- Median: 1
- Mode: 1
- Variance: 8.153
- Standard deviation: 2.855
- Skewed distribution? - Yes (positive)

Variable 5 (Single Epithelial Cell Size):

- Mean: 3.216
- Median: 2
- Mode: 2
- Variance: 4.903
- Standard deviation: 2.214
- Skewed distribution? - Yes (positive)

Variable 6 (Bare Nuclei):

- Mean: 3.545
- Median: 1
- Mode: 1
- Variance: 13.278
- Standard deviation: 3.644
- Skewed distribution? - Yes (positive)

Variable Descriptive Statistics Cont.

Variable 7 (Bland Chromatin):

- Mean: 3.438
- Median: 3
- Mode: 2
- Variance: 5.946
- Standard deviation: 2.438
- Skewed distribution? - Yes (positive)

Variable 8 (Normal Nucleoli):

- Mean: 2.867
- Median: 1
- Mode: 1
- Variance: 9.325
- Standard deviation: 3.054
- Skewed distribution? - Yes (positive)

Variable 9 (Mitoses):

- Mean: 1.589
- Median: 1
- Mode: 1
- Variance: 2.941
- Standard deviation: 1.715
- Skewed distribution? - Yes (positive)

Target/Dependent Variable

Our target variable is our class column because we are trying to find if it is benign or malignant and it's our only 2 outcomes.

Probability Questions

1. What is the probability that your target variable is greater than its mean?

Because Class is our target variable for our data set, it is a categorical variable consisting of benign and malignant. Therefore, it does not have a numerical mean. We used malignant as our target event and calculated the probability that a randomly selected observation is malignant. The probability of randomly selecting a malignant observation from our data set is 34.47%.

2. Choose an explanatory variable; Given that the target variable is greater than its mean, what is the probability that the chosen explanatory variable is ALSO greater than its mean?

For our explanatory variable, we chose V1 (clump thickness), which has a mean of 4.418 . Given that an observation is malignant, we calculated the probability that its clump thickness is ALSO greater than its mean. This means that 87.14% of the malignant observations have a clump thickness greater than the overall mean of 4.418.

Hypothesis Test Setup

We used the categorical target variable “Class” to subset the data into malignant and benign groups. Then we tested the difference between the two groups’ mean clump thickness.

H0: Mean of malignant ≤ Mean of benign

H1: Mean of malignant > Mean of benign

We used a 0.05 significance level on a one tailed two sample t tests assuming unequal variances.

Malignant group variance: 5.9

Benign group variance 2.8

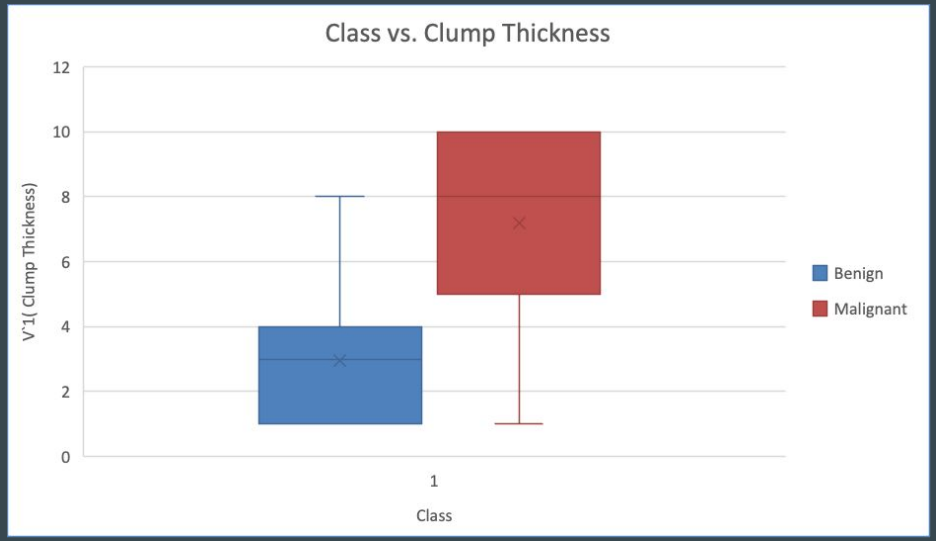
Hypothesis Test Results

	Malignant	Benign
N	241	458
Mean V1	7.2	2.96
Variance	5.9	2.8

Difference in means: 4.24
T-stat = 24.23
Critical t = 1.649
P-value = $3.78 * 10^{-78}$

Decision: t stat > critical and p value < 0.05 so we reject H_0.
The difference in means is statistically significant. Malignant biopsies have a higher mean clump thickness than benign biopsies, which makes clump thickness a strong indicator of our target variable.

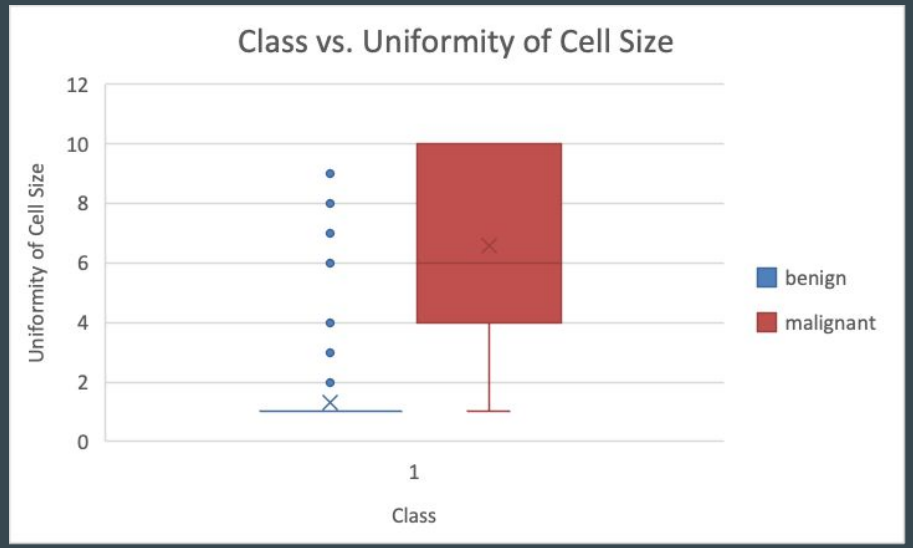
Explanatory Variable 1:



Relationship: Strong relationship, malignant tend to have higher clump thickness than benign. Clear difference between both classes.

Correlation Coefficient: 0.716

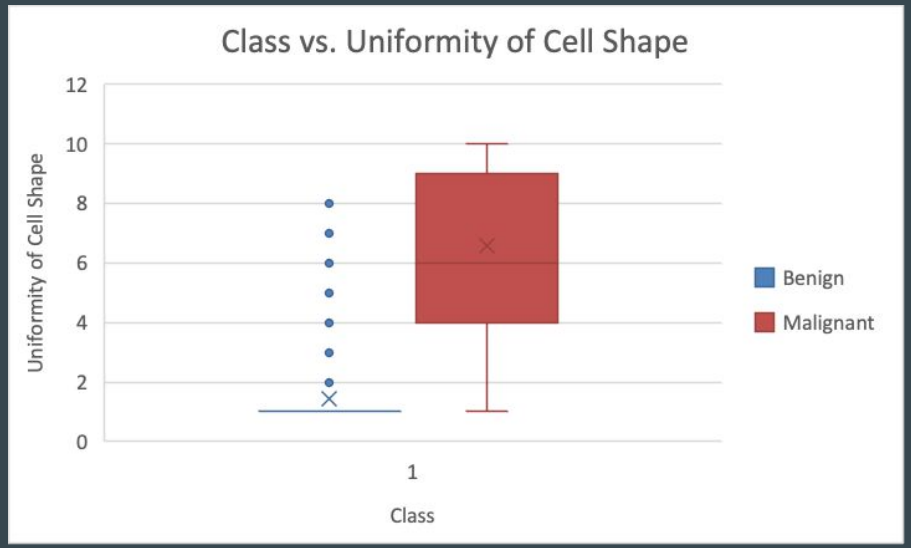
Explanatory Variable 2:



Relationship: Strong relationship, malignant tend to have higher values in cell size, benign is observed at lower values and has a few outliers.

Correlation Coefficient: 0.818

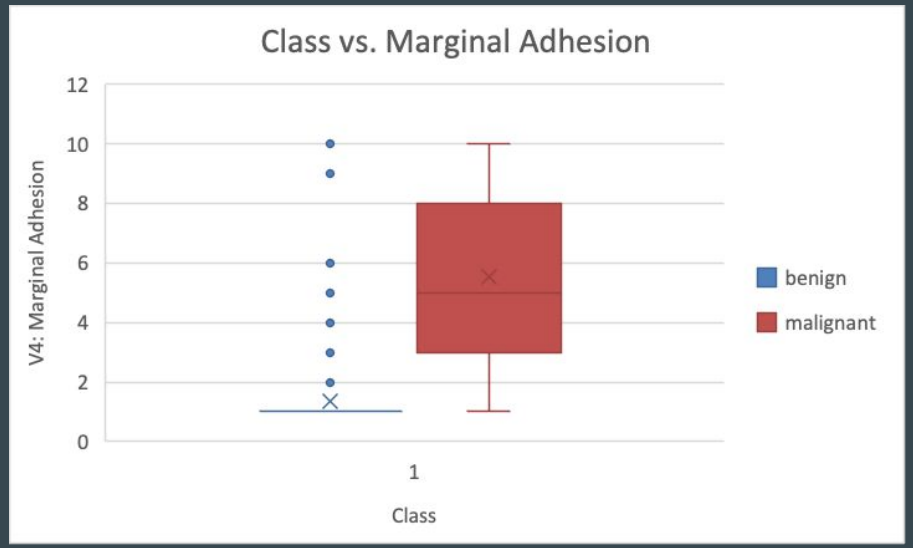
Explanatory Variable 3:



Relationship: Strong relationship, malignant observations have higher values in cell shape, benign is clustered towards the lower end.

Correlation Coefficient: 0.819

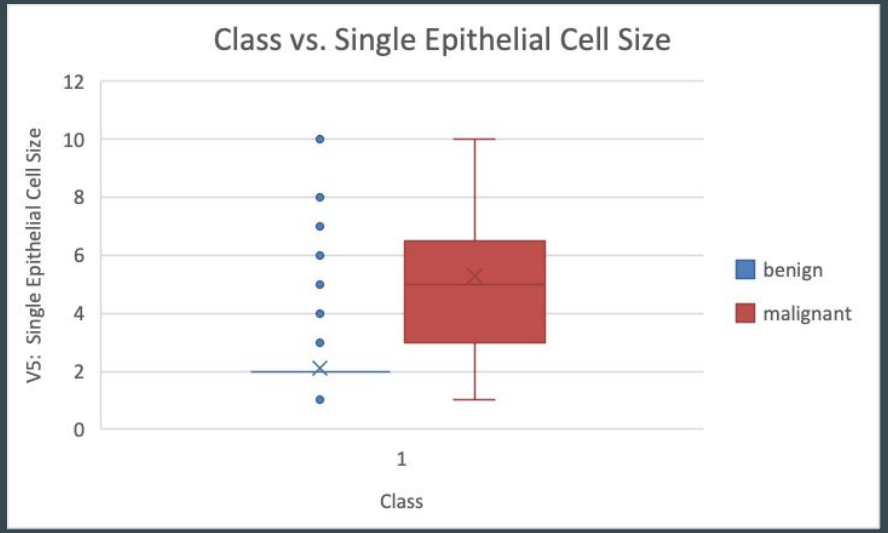
Explanatory Variable 4:



Relationship: Moderate relationship, malignant observations have higher marginal adhesion, some overlap between classes is observed.

Correlation Coefficient: 0.697

Explanatory Variable 5:



Relationship: Moderate relationship, malignant tend to have higher single epithelial cell size. Benign still observed on the lower end even though there are a few outliers.

Correlation Coefficient: 0.683

Correlation Coefficients and R squared

Variable:	Correlation Coefficient	R^2
Clump Thickness	0.716	0.51266
Uniformity Of Size	0.8179	0.66897
Uniformity Of Shape	0.81893	0.67065
Marginal Adhesion	0.6968	0.48553
Single epithelial size	0.68278	0.46619

R Squared shows the share of variation in class that lines up with each variable.

All five correlations are positive, higher values on each variable go with malignant biopsies. Cell size and cell shape are the strongest with about 67% of the variation in class lining up with each of them.

Marginal adhesion and single epithelial cell size are the weakest, this matches the overlap between benign and malignant in our charts. Correlation shows these variables move together with class, but it doesn't prove that one causes the other.

Multiple Linear Regression Predicting Breast Cancer Class

Our target variable is Class so for us to run the regression model it was in our best interest to make a separate column where **Benign = 0 and Malignant = 1.** Observations containing missing values were excluded from the regression.

Instead of looking at just 5 explanatory variables we chose all of them (V1 through V9) because later on we can explore to see which variables many or many not be significant in our data set.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.91832681
R Square	0.84332413
Adjusted R Sq	0.84122891
Standard Erro	0.19018384
Observations	683

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	9	131.025159	14.5583509	402.499146	4.456E-264
Residual	673	24.3423378	0.03616989		
Total	682	155.367496			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 99.0%*	*Upper 99.0%*
Intercept	-0.2476611	0.01642399	-15.079231	1.7659E-44	-0.2799095	-0.2154127	-0.2900868	-0.2052354
V1	0.03171309	0.00356404	8.89808147	5.211E-18	0.02471512	0.03871106	0.02250663	0.04091955
V2	0.02184501	0.0063729	3.42779497	0.00064557	0.00933184	0.03435818	0.00538281	0.03830721
V3	0.01563963	0.00623635	2.5078175	0.01238223	0.00339459	0.02788468	-0.0004698	0.03174909
V4	0.00824327	0.00399249	2.06469409	0.03933449	0.00040404	0.01608251	-0.0020699	0.01855649
V5	0.0100751	0.00523762	1.92360246	0.05482593	-0.0002089	0.02035915	-0.0034545	0.02360469
V6	0.04538628	0.00322084	14.0914237	1.0531E-39	0.03906217	0.05171039	0.03706634	0.05370622
V7	0.01917562	0.00504419	3.80152285	0.00015686	0.00927137	0.02907987	0.00614569	0.03220555
V8	0.01852933	0.00371989	4.98114721	8.0437E-07	0.01122534	0.02583333	0.00892028	0.02813839
V9	0.00097888	0.00496704	0.19707617	0.84382744	-0.0087739	0.01073163	-0.0118517	0.0138095

$$Class = \beta 0 + \beta 1V1 + \beta 2V2 + \ldots + \beta 9V9$$

Class = -0.248 + 0.032V1 + 0.022V2 + 0.016V3 + 0.008V4 + 0.010V5 + 0.045V6 + 0.019V7 + 0.019V8 + 0.001V9

Initial Model Results

The model has an **R-Squared** of approximately **0.843,** meaning that about **84.3% of the variation** in the coded Class variable is explained by V1-V9. The **Adjusted R-Squared** is approximately **0.841**, which indicates that the model still explains a large amount of variation after accounting for the number of predictors.

Most of the explanatory variables had statistically significant relationships with Class at a **significance level** of **0.05**. Because malignant was set to 1 and benign to 0, positive coefficients indicate that larger values of the explanatory variable are associated with the observation being more toward the malignant category.

Initial Model Results Continued ...

V9 had a p-value of approximately 0.844, making it the least statistically significant explanatory variable in the model. V5 was also slightly above the 0.05 significance level in the initial model with a p-value of approximately 0.055.

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 99.0%*	*Upper 99.0%*
Intercept	-0.2476611	0.01642399	-15.079231	1.7659E-44	-0.2799095	-0.2154127	-0.2900868	-0.2052354
V1	0.03171309	0.00356404	8.89808147	5.211E-18	0.02471512	0.03871106	0.02250663	0.04091955
V2	0.02184501	0.0063729	3.42779497	0.00064557	0.00933184	0.03435818	0.00538281	0.03830721
V3	0.01563963	0.00623635	2.5078175	0.01238223	0.00339459	0.02788468	-0.0004698	0.03174909
V4	0.00824327	0.00399249	2.06469409	0.03933449	0.00040404	0.01608251	-0.0020699	0.01855649
V5	0.0100751	0.00523762	1.92360246	0.05482593	-0.0002089	0.02035915	-0.0034545	0.02360469
V6	0.04538628	0.00322084	14.0914237	1.0531E-39	0.03906217	0.05171039	0.03706634	0.05370622
V7	0.01917562	0.00504419	3.80152285	0.00015686	0.00927137	0.02907987	0.00614569	0.03220555
V8	0.01852933	0.00371989	4.98114721	8.0437E-07	0.01122534	0.02583333	0.00892028	0.02813839
V9	0.00097888	0.00496704	0.19707617	0.84382744	-0.0087739	0.01073163	-0.0118517	0.0138095

Reduced Regression Model

We removed V9 because it had a p-value of approximately 0.844, which is substantially greater than our significance level of 0.05. This suggests that V9 does not provide meaningful additional explanatory power after accounting for the other variables.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.91832189
R Square	0.84331509
Adjusted R Sq	0.84145533
Standard Erro	0.19004818
Observations	683

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	8	131.023754	16.3779692	453.453334	2.067E-265
Residual	674	24.3437426	0.03611831		
Total	682	155.367496			

	Coefficients	*tandard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 99.0%*	*Upper 99.0%*
Intercept	-0.2472939	0.01630633	-15.165522	6.5096E-45	-0.2793112	-0.2152766	-0.2894155	-0.2051724
V1	0.03175904	0.00355387	8.93647339	3.8105E-18	0.02478106	0.03873702	0.02257889	0.04093919
V2	0.0219031	0.00636154	3.44304722	0.00061078	0.00941227	0.03439393	0.00547032	0.03833588
V3	0.01564319	0.00623188	2.51018957	0.0122998	0.00340697	0.02787942	-0.0004546	0.03174103
V4	0.00834665	0.00395506	2.11037379	0.03519401	0.00058093	0.01611237	-0.0018698	0.01856313
V5	0.01027938	0.00513036	2.00363757	0.04550869	0.00020597	0.0203528	-0.0029731	0.02353184
V6	0.04535532	0.00321471	14.1086587	8.5867E-40	0.03904326	0.05166738	0.03705125	0.05365938
V7	0.01908963	0.0050217	3.80142511	0.0001569	0.00922957	0.0289497	0.00611785	0.03206141
V8	0.01861865	0.00368955	5.04632528	5.7999E-07	0.01137426	0.02586304	0.00908802	0.02814928

$$Class = \beta 0 + \beta 1V1 + \beta 2V2 + \ldots + \beta 8V8$$

Reduced Regression Model Continued...

Removing V9 caused essentially **no change in R-Squared,** while Adjusted R-Squared slightly increased. Therefore, we consider the reduced model **better** because it is simpler while maintaining essentially the same explanatory power.

Also after removing V9, V5's p-value actually decreased to approximately **0.046**, making it statistically significant at the 0.05 significance level from **~0.055 to ~0.046.**

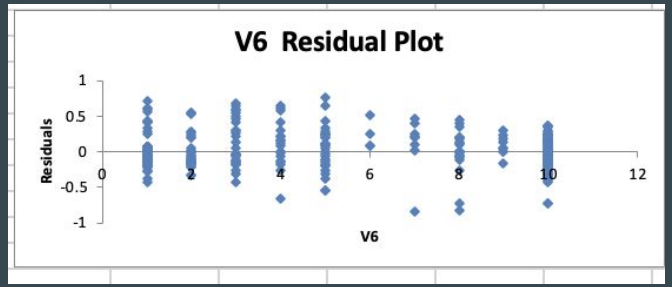
	Original	Without V9
R-Squared	~0.8433	~0.8433
Adjusted R-Squared	~0.8412	~0.8415
Predictors	9	8

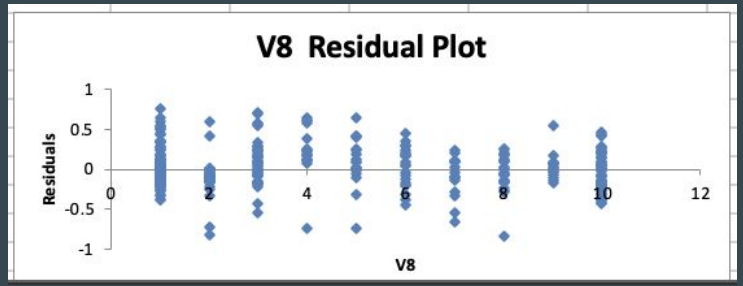
Does Our Model Meet the Assumptions?

Linearity: The residual plots are generally scattered around zero and do not show a strong curved pattern. Suggesting that the linearity assumption is reasonably satisfied.

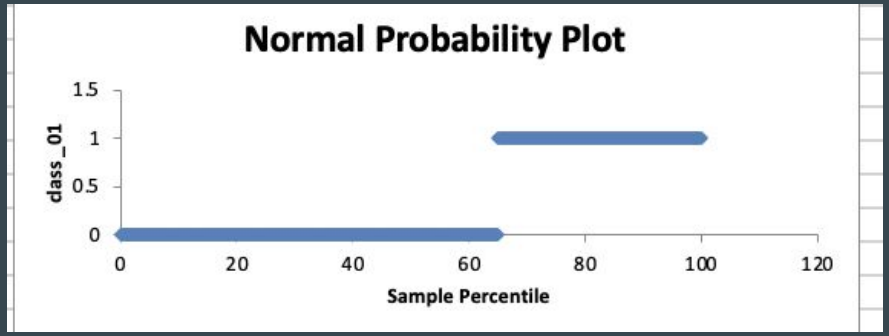
Independence: Each row represents a separate biopsy observation, so we assume the observations are independent.







Does Our Model Meet the Assumptions?



Normality: The normal probability plot does not show an approximately straight line. Instead, there are **two distinct horizontal groups at 0 and 1** because our target variable is binary. Therefore, the normality assumption is not satisfied well and better takes on the shape of a logistic function.

Overall, the model does not fully meet the assumption of linear regression. The biggest issue is that class is a **binary categorical variable** (benign = 0, malignant = 1), rather than a continuous response variable. Because of this using **logistic regression would be a much more appropriate way for predicting Class** rather than linear regression.